

# **SENG 310 ASSIGNMENT 5**

## **FINAL EVALUATION WITH USERS**

Team B03-07

University of Victoria  
SENG 310 – Human Computer Interaction

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# 1. STUDY DESCRIPTION

## 1.1 Prototype Overview and Design Requirements

The Palendar medium-fidelity prototype is an interactive web-based application designed to help friend groups coordinate social activities by visualizing shared availability across multiple schedules. The prototype addresses the core problem identified in PA1: the difficulty of finding common free times among friends with busy, conflicting schedules. As established through our PA2 user research with eight university students, this scheduling challenge causes frequent plan cancellations and reduces social interaction, negatively impacting mental health and social connection.

The prototype successfully implements all five "must be met" design requirements from PA2. First, users can create and view their personal calendar with color-coded categories for different life domains (academic, work, social), directly addressing Vedit's need for separate schedule components. Second, the Groups section enables participation in multiple friend groups simultaneously through a scrollable interface with group codes, solving Abheenaya's double-booking challenges. Third, clicking any group displays a dedicated group calendar showing aggregated member schedules, fulfilling Harshad's request for unified availability visualization. Fourth, green visual indicators automatically highlight time slots when all members are free, eliminating Apoorva's manual schedule comparison burden. Fifth, the Create Group functionality generates unique shareable codes for establishing new groups.

Beyond the core requirements, Palendar incorporates integrated chat functionality through the Friends tab, creating an end-to-end scheduling and communication solution. This design emerged from PA2 insights about fragmentation between scheduling tools and messaging platforms. The interface follows a three-button home dashboard architecture (Calendar, Profile, Events) implementing our PA3 "separation of concerns" approach, preventing information overload while maintaining direct access to primary functions. Design improvements from PA3's heuristic evaluation include enhanced visibility of system status through explicit confirmation messages, consistent green color semantics for availability indicators, reduced interface clutter through minimalist layouts, and persistent navigation elements supporting recognition over recall.

## 1.2 Evaluation Methodology and Participant Details

We conducted usability testing with four participants recruited from our university network, following the comprehensive evaluation plan developed in PA4. Each team member facilitated evaluation sessions, maintaining consistency through our standardized protocol script. The evaluation assessed whether users could successfully complete the core task workflow: login → navigate to groups → identify common free times → create hangout event.

**Participant 1** is a university student studying economics who regularly coordinates activities across multiple friend groups. She experiences frequent scheduling conflicts and currently uses basic calendar apps without group coordination features. Her technology comfort level is

high (4/5), making her representative of our target user base of tech-comfortable students struggling with group scheduling complexity rather than technology itself.

**Participant 2** is a computer science major at UVic who identified scheduling as the primary difficulty when planning time with friends, specifically "knowing what free time people have at what part of the day and what time during the week." This pain point aligns precisely with Palendar's value proposition, making this participant ideal for validating whether our solution addresses real user needs.

**Participant 3** is a university student who successfully navigated the interface and demonstrated strong understanding of the availability indicators. This participant provided valuable feedback on navigation element placement and visual hierarchy, contributing insights about first-time user expectations when locating key interface components.

**Participant 4** is a university student who articulated their thought process clearly throughout the evaluation, explaining their reasoning for each interaction. This participant's verbal protocol provided rich insights into user mental models and expectations regarding icon meanings and navigation patterns.

Each session followed our eight-phase protocol from PA4: pre-session setup, demographic interview, think-aloud training, task scenario presentation, task execution with observation, post-task interview, wrap-up, and immediate documentation. Sessions averaged 15-20 minutes. We used screen recording to capture interactions and audio recording to preserve think-aloud narration and interview responses. Each team member conducted evaluation sessions independently, ensuring diverse facilitation perspectives while maintaining protocol consistency. Both participants were asked to complete a realistic scenario: planning a movie night with their friend group by finding common availability and creating a hangout event. This scenario directly maps to the task descriptions developed in PA2, particularly addressing Apoorva's pain point that "hangout plans fail very frequently because people have different class times" and Harshad's need for clear visibility of when friends are free to prevent last-minute cancellations.

## 2. RESULTS

All four participants successfully completed all primary tasks without critical facilitator intervention, achieving a 100% task completion rate and strongly validating core interface usability. Task execution was efficient, with login taking under 30 seconds, group navigation ranging from 30-60 seconds, and event creation under 2 minutes per participant. The evaluation revealed consistent patterns in user interaction across all four sessions while also surfacing specific areas for improvement in navigation design, icon clarity, and information presentation. All findings are reported anonymously using participant identifiers (Participant 1-4) to protect privacy while preserving the richness of individual feedback.

**Task Completion and Core Functionality Validation:** All four participants successfully completed the core task flow without facilitator intervention, demonstrating strong interface usability. Participant 1 completed six distinct tasks including login, profile access, group joining using code "demo", group calendar viewing, event management, and friend chat. She reported finding "all the buttons as expected" and noted that "the buttons were where I would expect them to be," demonstrating effective adherence to user mental models. The facilitator observed: "Bro she was able to do all tasks without any intervention, a big W in my opinion," indicating exceptional learnability. Similarly, Participant 2 characterized the experience as "as straightforward as I can imagine," suggesting successful reduction of cognitive load. Participant 3 efficiently navigated through groups, correctly identified availability patterns, and successfully created events, even adjusting their selection when asked to find optimal availability (identifying the 28th with "10 on 10" availability). Participant 4 demonstrated clear understanding of the interface logic, correctly reasoning that the home page "is selecting our own availability" and that they needed to "go to groups to see when everyone's available." This consistent pattern of successful task completion across diverse users validates that Palendar's core workflow effectively implements PA2's task descriptions.

**Navigation and Information Architecture Insights:** While all participants successfully found the groups section, two participants (Participants 3 and 4) revealed important insights about navigation expectations. Participant 3 reported: "I didn't realize that the groups tab was down. Like here... I was trying to search for the groups thing here [at the top]. But then I saw it and then like obviously I figured it out." This indicates that bottom navigation placement, while ultimately discoverable, contradicts some users' initial expectations for primary navigation elements. The feedback suggests that critical features like Groups may benefit from more prominent placement or visual emphasis to match user mental models. However, the fact that participants quickly adapted ("obviously I figured it out") demonstrates that the interface supports rapid learning even when initial placement differs from expectations. Participant 4 navigated efficiently without this confusion, suggesting individual variation in navigation expectations rather than a universal usability barrier.

**Icon Clarity and Affordance Issues:** A consistent usability issue emerged regarding icon interpretation. Participant 4 clicked what they believed was a home or profile icon, which was actually the logout function. When asked what they expected: "I thought it would take me back to the home screen." Upon further discussion, they explained: "Because there's a person on

the icon, I would [think it goes to] profile... It would go back to home." This misinterpretation represents a violation of Nielsen's (1994) usability heuristic of recognition rather than recall—the icon's visual affordance (person symbol) suggests profile or account access rather than logout functionality. The facilitator acknowledged this issue, and Participant 4 emphasized: "We want the app to be really, first thing that you think of," highlighting the importance of intuitive icon design. This finding affects post-task workflows and could cause frustration if users accidentally log out when attempting to navigate home or access profile settings.

**Availability Visualization: Strengths and Limitations:** All four participants demonstrated understanding of the green availability indicators, with Participant 3 correctly identifying: "Because it says good availability, the green mark... on the 7th, 14th, 21st and 28th." When asked to clarify, they correctly reasoned: "Should be everyone" is available on those dates. Participant 4 similarly understood: "I'm sure the green is what indicates when everyone's free." This consistent interpretation validates the effectiveness of green as a semantic indicator for availability. However, Participant 2's earlier feedback that users "would also like to see when there is perhaps zero availability, medium availability" more prominently remains relevant—while the green indicators work well, additional granularity in availability visualization would enhance decision-making. Participant 3's interaction revealed an additional strength: they correctly identified today's date in the calendar through the blue highlighting ("The blue thing today, 23rd"), demonstrating that temporal orientation features function effectively.

**Visual Design and Aesthetic Considerations:** Participant 1's feedback on color scheme vibrancy ("the colour scheme could be fixed a little bit, maybe make it more vibrant") and nightmode suggestion remains the primary aesthetic concern identified. While functional performance was strong across all participants, enhanced visual appeal could improve user satisfaction and adoption. No other participants mentioned visual design issues, suggesting that while improvements would be beneficial, current aesthetics do not impede usability or task completion.

### 3. DISCUSSION

The evaluation results strongly validate Palendar's core value proposition while revealing specific opportunities for refinement in navigation design and icon affordances. The 100% task completion rate across four participants without critical intervention demonstrates that our implementation of PA2's design requirements successfully addresses real user needs. However, the feedback on navigation placement, icon clarity, and availability visualization highlights that meeting functional requirements is necessary but insufficient—interface elements must align with user mental models and minimize cognitive interpretation costs.

**Validation of Core Design Requirements and Task-Centered Approach:** The successful task completion by all four participants confirms that Palendar effectively implements the "must be met" requirements identified in PA2. Users could view group calendars, identify free times, create groups, join groups via codes, and manage personal schedules without critical difficulty. Participant 1's successful navigation through all six tested features (including the "should be met" chat functionality) demonstrates that our design scales across both core and secondary features. The consistency in positive feedback about workflow logic and button placement validates our PA3 decision to adopt a minimalist approach with clear affordances, supporting Norman's (2013) principles of discoverability and feedback in interface design. Participant 4's explicit articulation of their reasoning ("home page is selecting our own availability" so "I will be going to groups to see when everyone's available") demonstrates that the interface successfully communicates system structure and supports users in building accurate mental models.

**Navigation Hierarchy and Discoverability Challenges:** While participants ultimately located the groups section successfully, Participant 3's feedback reveals important insights about navigation expectations: "I didn't realize that the groups tab was down... I was trying to search for the groups thing here [at the top]." This finding suggests a mismatch between bottom navigation placement and user expectations for primary feature location. In our PA3 design rationale, we justified bottom navigation as standard practice for mobile-first interfaces. However, for a desktop or tablet web application, users may expect primary navigation at the top or in a sidebar, particularly for features as central as group management. This represents a platform-specific design consideration we should address. The quick recovery ("obviously I figured it out") indicates the issue affects initial discoverability rather than fundamental usability, but reducing time-to-first-action remains valuable for user experience. Potential solutions include: (1) adding visual emphasis to bottom navigation through color, size, or animation on first use, (2) implementing a brief tutorial highlighting navigation location, or (3) reconsidering navigation architecture to align with desktop web conventions while maintaining mobile responsiveness.

**Icon Affordance and Recognition Errors:** Participant 4's misinterpretation of the logout icon as a profile or home button reveals a critical affordance issue. The person icon visually suggests user account management (profile, settings) rather than session termination (logout). This violates Nielsen's (1994) heuristic of matching between system and the real world—common conventions associate person icons with profile access, not logout

functionality. Standard practice uses door/exit icons or explicit "logout" text for session termination. This misalignment could cause frustration through unintended logouts, particularly problematic if the application doesn't preserve user state. The participant's emphasis that "we want the app to be really, first thing that you think of" reinforces the importance of intuitive iconography. Recommended solutions include: (1) replacing the person icon with a conventional logout icon (door with arrow), (2) adding text labels to all navigation icons to reduce ambiguity, (3) moving profile access to a separate location (such as a dedicated profile icon) and using explicit signaling for logout, or (4) implementing a confirmation dialog for logout to prevent accidental session termination. This change would improve both discoverability (users can find profile/home when intended) and error prevention (users won't accidentally log out).

**Availability Visualization: Balancing Simplicity and Information Richness:** The consistent understanding of green availability indicators across all four participants (Participant 3: "the green mark" means "good availability," Participant 4: "the green is what indicates when everyone's free") validates our color semantic choices from PA3. However, Participant 2's earlier feedback about wanting to see "zero availability, medium availability" more prominently remains unaddressed and represents an opportunity for enhancement. Participant 3's interaction provides additional context—they successfully identified that certain days showed "10 on 10" availability versus "7 out of 10," demonstrating that when numerical indicators are present, users understand and value granular information. This suggests implementing a graduated visual system: different green intensities or additional color coding (green for all free, yellow for most free, red for few/none free) combined with numerical availability counts. This approach follows Shneiderman's (2003) visual information-seeking mantra: "overview first, zoom and filter, then details-on-demand." Users could quickly scan color-coded availability at a glance, then drill into specific days for detailed numerical breakdowns. This enhancement would strengthen our fulfillment of PA2's fourth design requirement ("group informs members about common free times") by making that information more immediately accessible and nuanced.

**Synthesis and Design Prioritization for Future Iterations:** The evaluation demonstrates that Palendar successfully addresses the core scheduling coordination problem but reveals three priority areas for refinement: (1) Navigation architecture—consider platform-appropriate placement of primary features or enhance discoverability of bottom navigation through visual emphasis, (2) Icon design—replace ambiguous person icon with conventional logout signaling and ensure all icons follow platform conventions, (3) Availability visualization—implement graduated color coding with numerical indicators to provide richer information at-a-glance. Secondary enhancements include color vibrancy improvements and nightmode implementation for aesthetic appeal. These refinements maintain the intuitive workflow and clear information architecture that participants praised while addressing specific friction points. The consistent 100% task completion rate indicates we have a strong foundation; these iterations would elevate the interface from usable to exceptional.

## 4. STUDY LIMITATIONS AND REFLECTIONS

**Prototype Fidelity Constraints:** As a medium-fidelity prototype, Palendar lacks full functionality that might reveal usability issues only apparent during extended use. For example, we could not evaluate how users manage calendar updates over time, handle notification preferences, or resolve scheduling conflicts when availability changes. The prototype does not implement actual calendar integration (like Google Calendar sync), which was a "should be met" requirement from PA2 that multiple interview participants requested. Real-world usage might reveal integration challenges or workflow friction that our simplified prototype cannot expose. Additionally, testing occurred in controlled settings rather than authentic contexts where users would naturally schedule hangouts, potentially affecting ecological validity of our findings. Longitudinal evaluation with users integrating Palendar into their actual daily scheduling practices would provide deeper insights into sustained usability and adoption patterns.

**Evaluation Process Insights and Facilitator Bias:** Conducting this usability evaluation provided valuable lessons about the importance of structured methodology and the challenges of maintaining neutral facilitation. Following our PA4 evaluation plan proved essential—having a detailed protocol prevented inconsistency between sessions and ensured we collected comparable data. However, we recognized the challenge of remaining neutral during testing; when Participant 2 raised the availability visualization issue, the facilitator's immediate acknowledgment ("that was something we encountered as well") may have influenced subsequent feedback or made the participant feel obligated to continue in a critical vein. More neutral probing ("Can you tell me more about what you'd expect to see?") would have elicited richer details without suggesting the team already recognized the problem. This experience highlighted that facilitator responses, even when well-intentioned, can introduce bias into qualitative data collection. Future evaluations would benefit from more rigorous facilitator training and possibly using team members unknown to participants to minimize social desirability effects.

**Lessons Learned and Reflections on the Process:** This evaluation reinforced that even successful task completion does not guarantee optimal user experience. Both participants succeeded at all tasks, yet their feedback revealed meaningful opportunities for improvement in visual design and information presentation. This highlights the importance of qualitative feedback collection beyond quantitative metrics—satisfaction and efficiency matter as much as task completion. We learned that "straightforward" interfaces can still benefit from refinement; Participant 2's characterization of the interface as "straightforward as I can imagine" did not prevent him from identifying the availability visualization limitation, demonstrating that users can recognize enhancement opportunities even in systems they find usable. The think-aloud protocol proved invaluable for understanding user reasoning, though we observed that some participants found it challenging to verbalize their thoughts while concentrating on tasks. The process taught us that usability testing requires patience—allowing silence and avoiding premature intervention yields more authentic insights into natural user behavior. Most importantly, we learned that user feedback should directly drive design iteration; the specific, actionable suggestions from both participants (availability



heat map, color vibrancy, nightmode) provide a clear roadmap for our next prototype iteration, demonstrating the tangible value of user-centered design methodologies.

## REFERENCES

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